Cian Roche

cianmroche.github.io | roche@mit.edu | 0000-0002-3400-6991

Education

Massachusetts Institute of Technology 2021 – 2027 (expected)
Ph.D. Candidate, Physics
Thesis: *"The Dark Universe Through The Strong Lens: Cluster and Galaxy-Scale Probes of the Dark Sector"*
Advisors: Prof. Michael McDonald, Prof. Mark Vogelsberger

Universität Tübingen 2019 – 2021
M.Sc., Mathematical Physics, 1.1 GPA[†]
Thesis: *"Exact parallel waves in general relativity"*
Advisor: Prof. Carla Cederbaum

University College Cork 2015 – 2019
B.Sc., Physics, 1st class honours

Honors and Awards

MIT Physics Henry W. Kendall Teaching Fellowship 2026
DAAD Graduate Program Full Scholarship 2019
University College Cork College Scholar 2019
Max Planck Institute for Plasma Physics Research Scholarship 2018
Naughton Research Fellowship (one of 8 in Ireland) 2017
Naughton Foundation Scholarship (one of 29 in Ireland) 2015
All-Ireland Scholarship (one of 125 in Ireland; declined) 2015

Publications

ADS library: https://tinyurl.com/CianRocheADS 72 first-author citations, 162 citations, h-index 7

First Author or Significant Contribution

8. Dutra, I., Natarajan, P., **Roche, C.** 2026; *"Where Cluster Galaxies Reside and How Strongly They Lens"*, *submitted, ApJ*, arXiv:2610.14167
7. **Roche, C.**, McDonald, M., Natarajan, P., Sharon, K., Dutra, I., Meneghetti, M., Vogelsberger, M. 2026; *"Host-Dependence and Line-of-Sight Effects on Galaxy-Galaxy Strong Lensing in Clusters"*, *submitted, ApJ*, arXiv:2608.30079
6. Fosdick, K., McDonald, M., Khullar, G., Gladders, M., Häußler, B., **Roche, C.**, et al. 2026; *"Quantifying the Dominance and Emergence of Brightest Cluster Galaxies In SPT and Planck Clusters: A Novel Application of the Gini Index"*, *accepted, ApJ*
5. **Roche, C.**, Vogelsberger, M., McDonald, M., Dutra, I., Natarajan, P., Shen, X., Shreeram, S., Metcalf, R., Sharon, K., Birrer, S., Lee, W., Meneghetti, M. 2026; *"A Consistent Implementation of Cluster Strong Lensing in Cosmological Simulation Light Cones"*, OJAp, 9, arXiv:2605.30433
4. Tran, V., Shen, X., Vogelsberger, M., Gilman, D., O'Neil, S., **Roche, C.**, Zier, O., Gao, J. 2024; *"Novel Density Profile for Isothermal Cores of Dark Matter Halos"*, Phys. Rev. D 113, 063024, arXiv:2411.11945
3. **Roche, C.**, McDonald, M., Borrow, J., Vogelsberger, M., Shen, X., Springel, V., Hernquist, L., Pakmor, R., Bose, S., Kannan, R. 2024; *"Brightest Cluster Galaxy Offsets in Cold Dark Matter"*, OJAp, 7, arXiv:2402.00928
2. **Roche, C.**, Necib, L., Lin, T., Ou, X., Nguyen, T. 2024; *"The Escape Velocity Profile of the Milky Way from Gaia DR3"*, ApJ, 972, 70, arXiv:2402.00108

[†]on a scale of 1-5, where 1 is the highest grade.

1. **Roche, C.**, Aazami, A.B. & Cederbaum, C. 2023; *"Exact parallel waves in general relativity"* Gen Relativ Gravit 55, 40, doi:10.1007/s10714-023-03083-x

Collaboration Papers

4. Lee, W., et al. (incl. **Roche, C.**) 2026; *"TNG-Cluster SIDM: Evolution of massive cluster mergers under the self-interacting dark matter paradigm"*, *submitted, ApJ*, arXiv:2610.18957
3. Rose J. C., et al. (incl. **Roche, C.**) 2025; *"The DREAMS Project: A New Suite of 1,024 Simulations to Contextualize the Milky Way and Assess Physics Uncertainties"*, arXiv:2512.00148
2. Nguyen, T., et al. (incl. **Roche, C.**) 2024; *"How DREAMS are made: Emulating Satellite Galaxy and Subhalo Populations with Diffusion Models and Point Clouds"*, ApJ, 997, 336, arXiv:2409.02980
1. Rose J. C., et al. (incl. **Roche, C.**) 2024; *"Introducing the DREAMS Project: DaRk mattEr and Astrophysics with Machine learning and Simulations"*, ApJ, 982, 68, arXiv:2405.00766

Non-Refereed

2. **Roche, C.** & Necib, L. 2023; "*Gaia DR3 6D Kinematics*" Zenodo doi:10.5281/zenodo.8088365
1. **Roche, C.** & Garnavich, P. 2020; "*Testing Progenitor Models Using the Late-Time Light Curve of Supernova 1992A*" Res. Notes AAS 4, 207, doi:10.3847/2515-5172/abcb05

Teaching

Massachusetts Institute of Technology

Average rating by students 6.95/7 — highest-rated[‡] across 55 semesters of recorded 8.13/8.14 evaluations.

Teaching Assistant, Experimental Physics II (8.14) ..... Spring 2026
Teaching Assistant, Experimental Physics I (8.13) ..... Fall 2024
Teaching Assistant, Experimental Physics I (8.13) ..... Spring 2024
Teaching Assistant, Experimental Physics I (8.13) ..... Fall 2023
Teaching Assistant, Experimental Physics I (8.13) ..... Fall 2022
Instructor, Gaia DR3 Hackathon ..... June 2022

University College Cork

Teaching Assistant, Introductory Physics II ..... Spring 2018
Leader, Peer Assisted Learning Scheme ..... 2016 – 2018

Presentations

Summary: *2 guest lectures, 12 invited talks (1 scheduled), 7 conference contributions (1 scheduled).*

Guest Lectures

Experimental Physics I – Massachusetts Institute of Technology — Oct 2022
Graduate Mathematical Physics – Universität Tübingen — July 2021

Invited (Colloquia/Seminars)

Talk "*Calibrating Strong Lenses for Precision Dark Sector Science*" — Dec 2026
Harvard University

Talk "*Are Cluster Galaxies Too Good at Lensing?*" — Oct 2026
MIT Kavli Institute for Astrophysics and Space Research

Talk "*Are Cluster Galaxies Too Good at Lensing?*" — Oct 2026
Stanford University

Talk "*Illuminating the Dark Sector with Strong Gravitational Lensing*" — Sep 2026
Stony Brook University

[‡]Among teaching staff with a minimum of 3 evaluations.

Talk "*Illuminating the Dark Sector with Strong Gravitational Lensing*" Sep 2026
University of Michigan, Dept. of Astronomy

Talk "*Illuminating the Dark Sector with Strong Gravitational Lensing*" Sep 2026
University of Michigan, Dept. of Physics

Talk "*Light Cone Strong Lensing in Cosmological Hydrodynamical Simulations*" Mar 2026
MIT Kavli Institute for Astrophysics and Space Research

Talk "*Understanding Dark Matter with Simulated Strong Lensing*" Sep 2025
Yale University

Talk "*Dark Matter Self-Interaction via Galaxy Cluster Dynamics*" Feb 2024
Harvard University

Talk "*Galaxies Lacking Dark Matter Produced in a Cosmological Simulation*" Apr 2022
MIT Kavli Institute for Astrophysics and Space Research

Talk "*Exact parallel waves in general relativity*" June 2021
Mathematical Physics Colloquium – Universität Tübingen

Talk "*Hermiticity of Young Operators*" June 2020
Young Tableaux Seminar Series – Universität Tübingen

Contributed

Talk "*The Dark Universe Through the Strong Lens: Cluster- and Galaxy-Scale Probes of the Dark Sector" (dissertation talk, submitted)* Jan 2027
American Astronomical Society Meeting 249 – Salt Lake City

Poster "*Are Cluster Galaxies Too Good at Lensing?*" Sep 2026
Probe Combination for Dark Matter Physics in the Era of Large Surveys – Kavli Institute for Cosmological Physics, University of Chicago

Poster "*The Line of Sight as a Major Systematic of Cluster Strong Lensing*" July 2026
Gravitational Lensing at the Turning Point – Sexten Center for Astrophysics

Talk "*The Observation-Simulation Connection and the Line of Sight in Cluster Strong Lensing*" June 2026
Strong Gravitational Lensing Workshop – University of Montpellier

Talk "*Consistent Strong Lensing Predictions from Cosmological Simulations*" Oct 2025
Scaling up Lensing Workshop – University of Liège

Poster "*Unveiling Dark Matter: Testing CDM with the First End-to-End Cluster Strong Lensing Simulations*" July 2025
Tracing Cosmic Evolution with Galaxy Clusters V – Sexten Center for Astrophysics

Talk "*Testing CDM with the First End-to-End Cluster Strong Lensing Simulations*" June 2025
European Astronomical Society Meeting 2025 – University College Cork

Mentorship and Outreach

Mentor, Physics Directed Reading Program, MIT 2026
Member, MIT Astrogazers ..... 2024 – Present

MIT GAGA Initiative Co-chair 2022 – 2023 | Organizing Committee 2021 – 2024
The Graduates Advising Graduate Admissions program offers information and mentorship to underrepresented groups in physics, and advises on improvements to the admissions process.

MIT PhysGAAP Program 2021 – 2023
The Physics Graduate Application Assistance Program provides guidance to prospective applicants, and liaises with physics department leadership on issues of accessibility in the graduate application. I have acted as one of the primary organizers and have mentored 5 students through the program.

Student Academic Representative, University College Cork ..... 2017 – 2019

Leadership and Service

Admissions Committee Member, MIT Physics Graduate Program ..... 2023 Cycle, 2024 Cycle
Reviewed ~ 300 *applications to the MIT physics graduate program.*

Admissions Committee Member, MIT Summer Research Program ..... 2023 Cycle
Volunteer, AAS Meeting 241 Graduate School Fair 2023
Graduate Student Office Manager, MIT Kavli Institute ..... 2022 – 2025
Organizer, MIT Kavli Institute Graduate Student Lunch Series ..... 2022 – 2023
Member, MIT Physics Graduate Student Council ..... 2021 – 2023
Organizing Committee, MIT Graduate Student Union ..... 2021 – 2023
Astronomy Officer, UCC Physics and Astronomy Society ..... 2017 – 2019

Educational Resources

These are some of the educational resources and tools I have produced to support my peers, my students or the community at large.

figinvert https://github.com/CianMRoche/figinvert
A lightweight command line tool based in python that inverts vector and raster graphics for dark mode talks. In the default mode, mimics Overleaf's PDF inversion.

Caustica https://cianmroche.github.io/assets/caustica/
An educational and intuition-building online strong lensing simulator, also capable of producing high-quality lensing media such as video and vector graphics.

χ by Eye https://cianmroche.github.io/assets/chi_by_eye/
A learning tool in a GeoGuessr-like format for training students to recognize believable or unrealistic statistics in a plot by eye. Sandbox mode is particularly useful for gaining hands-on intuition.

AstroWiki https://publish.obsidian.md/astrowiki/
I created and now co-maintain a set of living notes covering most of graduate level astrophysics, specifically designed to aid the next generations of graduate students when studying for the MIT astrophysics oral qualification exam.